AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER

TEST REPORT NO. 2013 - 19

Mc CORMICK FOUR-WHEEL TRACTOR
Model : C100max

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Mc CORMICK c100max FOUR-WHEEL TRACTOR

Test Report No	:	2013 - 19
Date of Test	:	January 3, February 5 & 22, 2013
Test Valid Until	:	February 22, 2018
Brand	:	Mc CORMICK
Model	:	C100max
Serial No.	:	LU5CP39115
Type	:	Four-Wheel Drive Tractor
Manufacturer	:	ARGO TRACTORS S.P.A. 42042 Fabbrico (RE) Italia Via G. Matteotti, Italy
Test Requested by	:	SUPER TRADE MACHINERIES GLOBAL, INC. 937 EDSA Philam, Quezon City
Machine Classification	:	Commercial

SUMMARY

The Mc CORMICK C100max Four-Wheel tractor was tested to establish its performance. It was tested at the original settings set by the manufacturer.

The Maximum Power developed at the Power-Take-Off (PTO) shaft of the tractor was 61.5 kW at 601 rpm PTO Shaft Speed. The Engine Crankshaft Speed at this point was 2176 rpm with Fuel Consumption Rate of 17.9 L/h. The PTO Shaft Torque at this point was 99.6 kg-m. The Maximum PTO Shaft Torque was 125.3 kg-m which occurred at 411 rpm PTO Shaft Speed.

The Radii of Turning Circle of the tractor when making a right turn were 5863 mm without brake and 4188 mm when brake was applied on the right rear wheel. On the other hand, the Radii of Turning Circle of the tractor when making a left turn were 5663 mm without brake and 4200 mm when brake was applied on the left rear wheel.

The Radii of Turning Area of the tractor when making a right turn were 5958 mm without brake and 4283 mm when brake was applied on the right rear wheel. On the other hand, the Radii of Turning Area of the tractor when making a left turn were 5758 mm without brake and 4295 mm when brake was applied on the left rear wheel.

The Travelling Speeds of the tractor at the engine full throttle speed setting (Pre-set) recommended by the manufacturer/Applicant of test at different gear settings were the following: At FORWARD shuttle setting, at "LOW" speed range the travelling Speeds were 1.89 kph, 2.97 kph, 3.74 kph and 5.55 kph at 1st, 2nd, 3rd and 4th gear settings, respectively, at "MEDIUM" speed range the travelling Speeds were 4.51 kph, 7.07 kph, 8.93 kph and 13.0 kph at 1st, 2nd, 3rd and 4th gear settings, respectively, and at "HIGH" speed range the travelling speeds were 11.1 kph, 17.3 kph, 21.8 kph and 31.4 kph at 1st, 2nd, 3rd and 4th gear settings, respectively.

At REVERSE shuttle setting, at "LOW" speed range the travelling Speeds were 1.95 kph, 3.02 kph, 3.80 kph and 5.59 kph at 1st, 2nd, 3rd and 4th gear settings, respectively, at "MEDIUM" speed range the travelling Speeds were 4.58 kph, 6.63 kph, 8.89 kph and 13.2 kph at 1st, 2nd, 3rd and 4th gear settings, respectively, and at "HIGH" speed range the travelling speeds were 11.2 kph, 17.9 kph, 22.1 kph and 32.8 kph at 1st, 2nd, 3rd and 4th gear settings, respectively.

SCOPE OF TEST

The tractor was tested to establish its performance characteristics. The test was conducted on January 3, February 5 & 22, 2013 at the AMTEC Test Laboratory, UPLB, College, Laguna. It was evaluated on the following items:

1. Suitability of operator's manual;
2. Verification of Specifications
3. Laboratory tests
    a. PTO load performance test;
    b. Determination of Turning Circle
    c. Determination of Turning Area
    d. Determination of Travelling Speeds

METHOD OF TEST

The tractor was tested based on Philippine Agricultural Engineering Standards, PAES 119:2001 Agricultural Machinery – Four-Wheel Tractor – Methods of Test.

DESCRIPTION OF THE TRACTOR

The Mc Cormick C100max Four-Wheel Tractor (**Figure 1**) was a 69.0 kW (92.5 Hp) four-wheel drive tractor.

The tractor used a four-stroke cycle, water-cooled, turbocharged, Intercooled, four cylinder diesel engine. Fuel feed system was a force-feed type using a circulation pump. The fuel tank (**Figure 2**) was located just beneath the left ladder of the tractor. It utilized a dry dual type air cleaner (**Figure 3**). Lubrication system was force-feed lubricating type. Cooling was done through circulation of water from cylinder jacket to the radiator using a circulation pump. It was started by an electric starting system. A vertical muffler was used in the exhaust system.

The tractor's transmission system had 12 forward speeds and 12 reverse speeds. The speed settings were grouped into SHUTTLE (forward and reverse), RANGE (Low, Medium and High), and Main gear settings (1^{st}, 2^{nd}, 3^{rd} and 4^{th}). The Main gear was fully synchronized while the range gear could only be shifted when the tractor was completely stopped. The range and main gear settings could be set through a lever both located at the right side of the operator's seat. The Shuttle lever was located on the left side and beneath the steering wheel. A parking brake was also provided and can be actuated through a lever on the left side of the operator's seat .

The tractor's direction was being controlled through a hydrostatic power steering system with the steering wheel (**Figure 4**) located in front of the driver's seat. The brakes of the rear wheels were independent of each wheel, however, they could be locked-on through the pedals. A differential lock could be set by a push button switch located underneath the operator's seat. The size of the rear wheel tires were 540/70 R 34 or 54.0 cm x 86.4 cm (21.2 in x 34 in) and the front wheel tires were 440/65 R24 or 44.0 cm x 61.0 cm (17.3 in x 24.0 in). Both have spikes.

The tractor had a three-point hitch linkage (**Figure 5**) for mounted implements and a swinging drawbar (**Figure 6a**) for pulling a trailer or any trailing implement.

The Power-Take-Off shaft (**Figure 6b**) of the tractor was located at the rear of the tractor and was set at 540 rpm. The PTO shaft could be engaged/disengaged through a lever located at the left side of the operator seat and using a hand lever clutch located in front of the operator's seat underneath the steering wheel.

The tractor had two headlights, two brake lights, four turning lights (two in front and two at the back), two fog lights, one working light, and one blinker.

The operator's seat (**Figure 7**) was adjustable to fit different sizes of operators. It was provided with safety seat belt.

The tractor controls (**Figure 8**) were located in front of and beside the operator's seat and included the following: steering wheel, clutch pedal, hand throttle lever, ignition key, main gear shift lever, Shuttle lever, range gear shift lever, differential lock switch, brake pedals, foot throttle pedal, PTO controls, hydraulic lift draft control lever, hydraulic lift position control lever and auxiliary hydraulic control lever. The panel board (**Figure 8**) which was also located in front of the operator's seat contained the following: tachometer, fuel gage, oil temperature, hour meter, warning lights (engine coolant high temperature, battery charging indicator, oil pressure warning light, air filter blockage warning light, parking brake light and brake fluid low level warning light), and operation indicator lights (turning lights, main beam, sidelights indicator, fuel reserve indicator, PTO engaged, four wheel drive indicator, and differential lock indicator). **Table 1** shows the detailed specifications of the tractor.

Figure 1. Mc CORMICK C100max Four-Wheel Tractor

Figure 2. Fuel tank

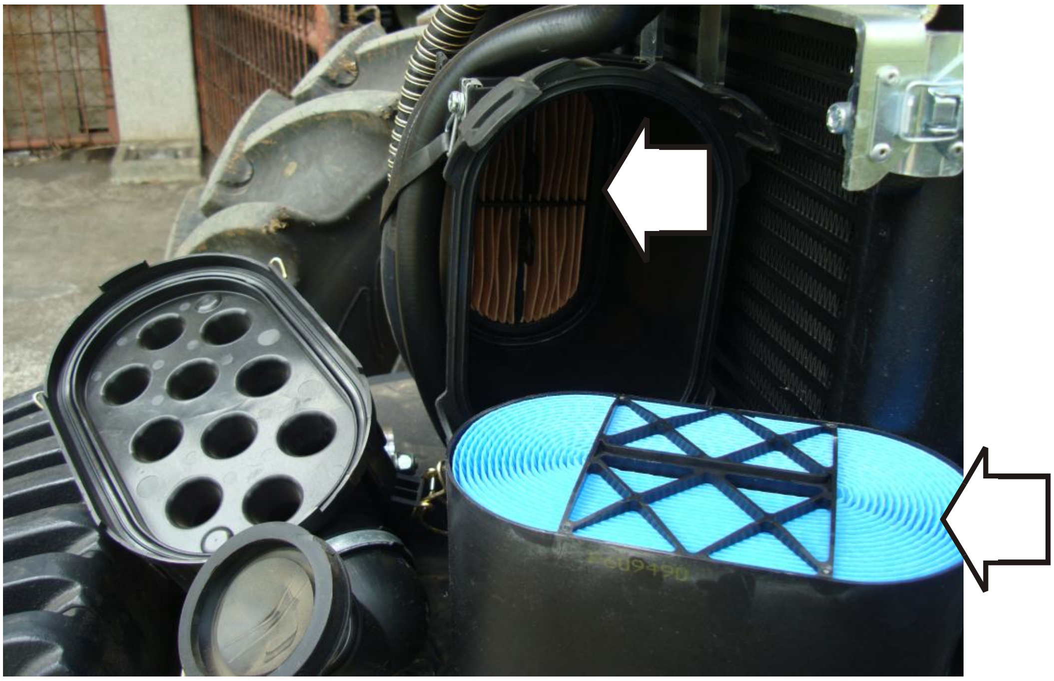

Figure 3. Air Cleaner

Figure 4. Steering wheel

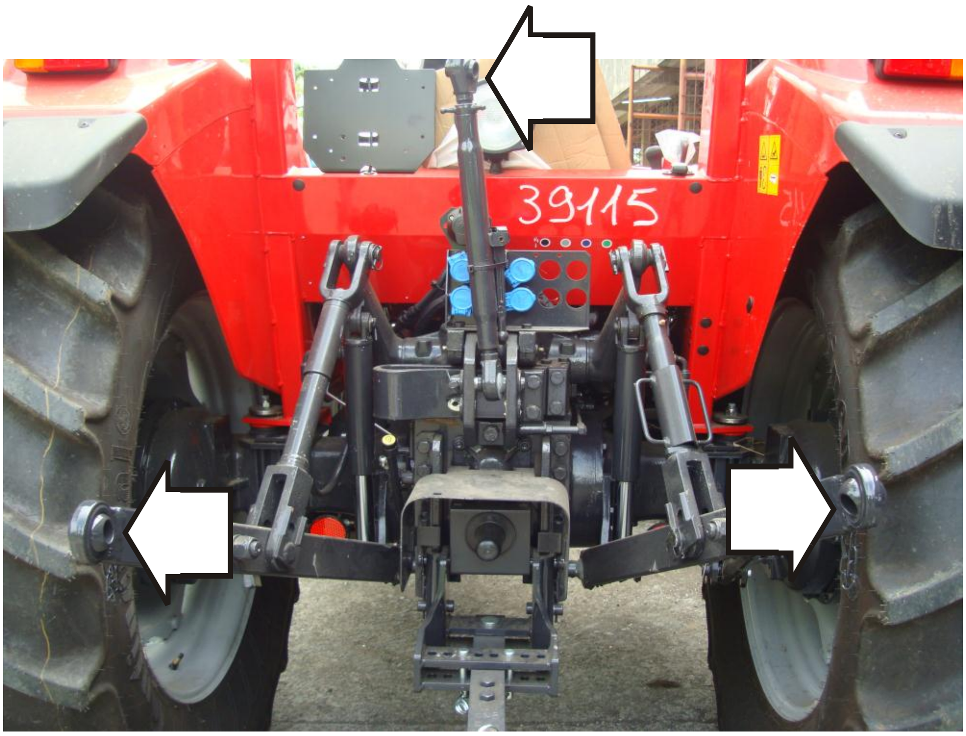

Figure 5. Three-Point Hitch

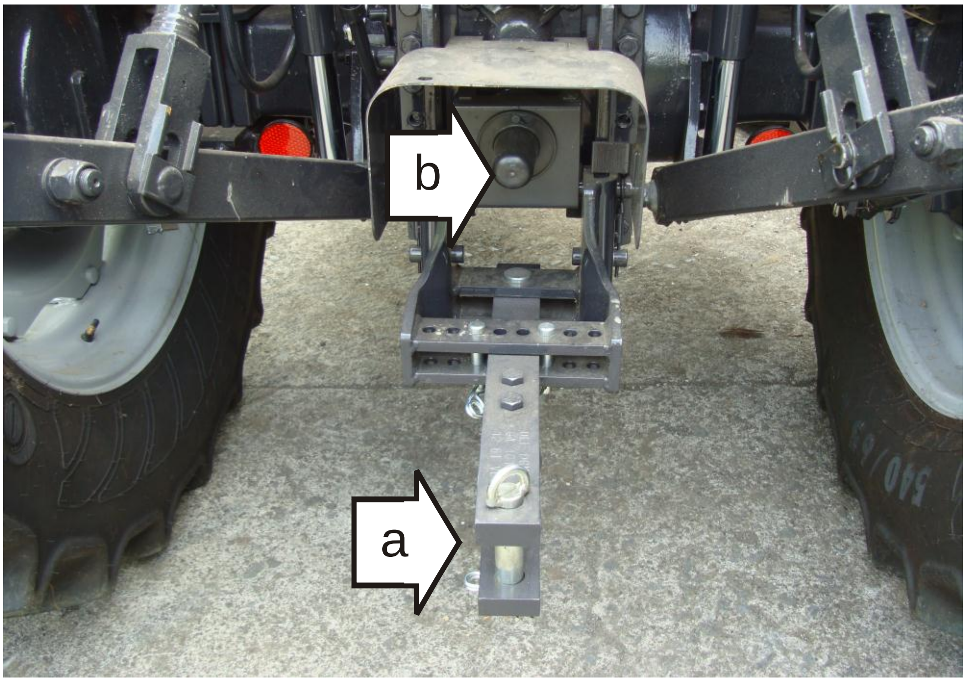

Figure 6. (a) Swinging drawbar
(b) PTO output shaft

Figure 7. Operator's seat

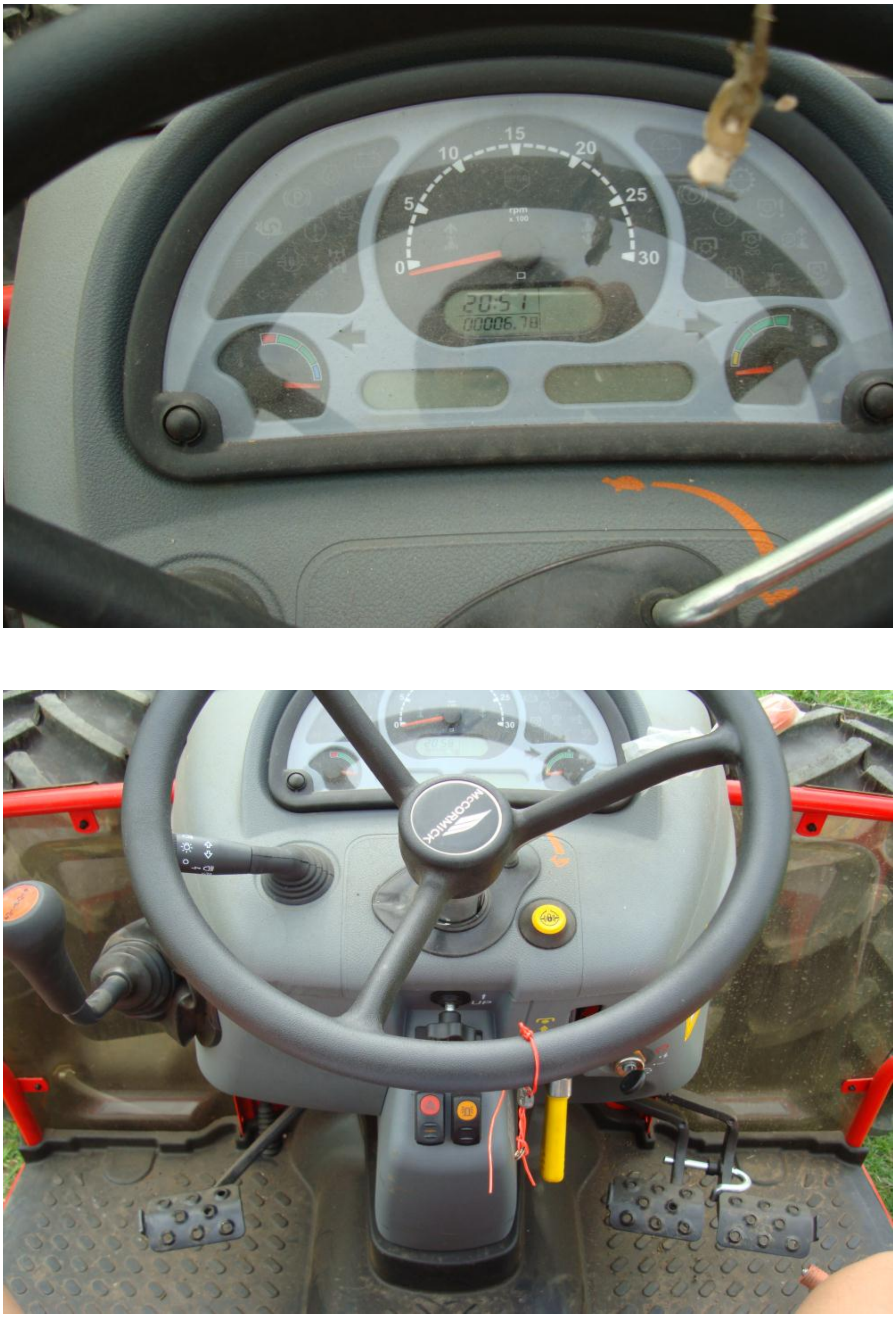

Figure 8. Tractor’s control and panel board

SPECIFICATIONS

Table 1. Detailed specifications of the Mc CORMICK C100max Four-Wheel Tractor

1.	Tractor specifications (Based on actual ocular inspection of the tractor and operator's manual submitted)			
	1.1	Brand	:	Mc CORMICK
	1.2	Model	:	C100max
	1.3	Serial No.	:	LU5CP39115
	1.4	Make	:	Italy
	1.5	Type	:	Four-Wheel Drive
	1.6	Rated Maximum Power (kW)	:	69.0 (Based on Operator's manual submitted)
2.	Dimensions and Weight			
	2.1	Overall length (mm) With front ballast	:	4125
	2.2	Overall width (mm)	:	2120
	2.3	Overall height (mm)	:	2570 (ROPS)
3.	Tractor's engine			
	3.1	Brand	:	Perkins
	3.2	Type	:	Four-Stroke Cycle, Water-cooled, Turbocharged, Intercooler, four Cylinder Diesel Engine
	3.3	Model	:	1104A-44A
	3.4	Make	:	England
	3.5	Serial Number	:	U180191W
	3.6	Rated Output Power (kW)	:	69.0
	3.7	Number of Cylinder	:	4

Table 1. Con't

	3.7	Bore x stroke (mm)	:	105 x 127
	3.8	Total displacement volume (cc)	:	4400
	3.9	Fuel system		
		3.9.1 Type of fuel feed	:	Circulation pump
		3.9.2 Fuel used	:	Diesel
	3.10	Cooling system	:	Water-cooled
	3.11	Lubricating system	:	Force-feed type
	3.12	Starting system	:	Electric type using battery
	3.13	Air cleaner	:	Dry Dual type with primary and secondary elements
	3.14	Exhaust system	:	Vertical muffler
4.	Power-Take-off			
	4.1	Location	:	At the rear of the tractor
	4.2	Diameter of PTO shaft end (mm)	:	35.0
	4.3	Number of splines	:	6
	4.4	Height above ground (mm)	:	745
	4.5	Rated Shaft Speed(rpm)	:	540
	4.6	Direction of rotation (viewed from the rear of the tractor)	:	Clockwise
	4.7	Mode of Operation	:	Lever
5.	Drawbar			
	5.1	Type	:	Swinging
	5.2	Diameter of hitchpoint (mm)	:	28

Table 1. Con’t

6.	Tire				
	6.1	Front			
		6.1.1	Number	:	2
		6.1.2	Size (cm)	:	44.0 x 61.0 440/65 R24
		6.1.3	Track width (mm)	:	1715
	6.2	Rear			
		6.2.1	Number	:	2
		6.2.2	Size (mm)	:	54 x 86.4 (540/65 R 34)
		6.2.3	Track width (mm)	:	1600
7.	Wheel base (mm)			:	2325
8.	Ground Clearance (mm)			:	375 (Drawbar bracket)

Table 2. Dimension of linkage geometry (see also **Figure 9**)

Particulars		Data (mm)
Length of lift arms	A	275
Length of lower links	B	950
Horizontal distance between two lower link points	u	385
Horizontal distance between the lift arm end points	v	530
Length of upper link	S	590-970
Distance of lower link points to lift rod pivot points on lower links	D	500
Length of lift rods	L	580-770
Height of lower hitch points relative to the rear-wheel axis		
Low position	h	325
High position	H	255
Height above ground of lower hitch points when locked in transport position		980
Three-point hitch		CATEGORY II

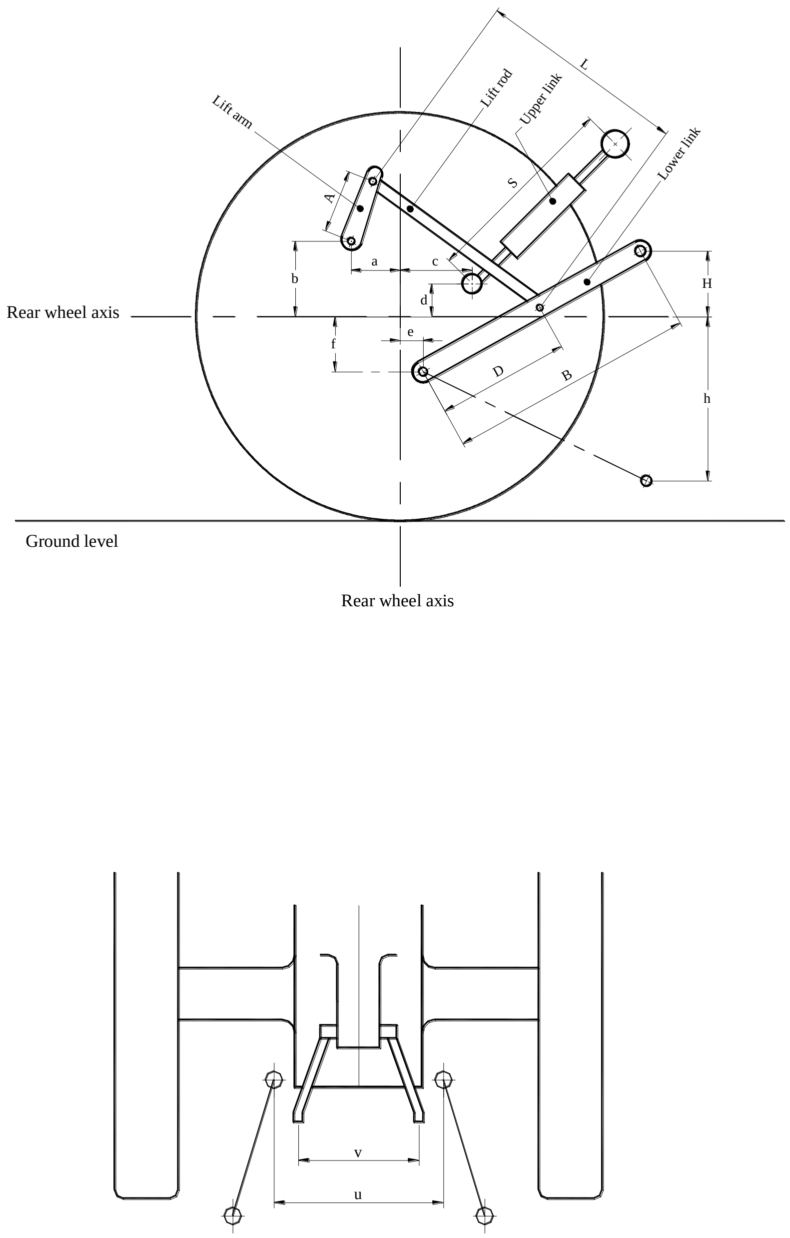

Figure 9. Linkage Geometry

OPERATOR'S MANUAL

The operator's manual submitted with the tractor was for Cmax Series Mc CORMICK tractors: C60, C75, C85, C95, and C105. It was presented in English and contained the following: Tractor identification, Introduction, Warranty and Safety notes, Identification of tractor parts, instrumentations and controls, Detailed pre-operation and operation procedures, General maintenance, inspection and service procedures, and List of technical specifications.

RESULTS AND DISCUSSION

The Mc CORMICK C100max Four-Wheel tractor was tested to establish its performance.

A. DETERMINATION OF TURNING CIRCLE AND TURNING AREA

The tractor was tested in a suitable concrete area. It was tested by operating the tractor at the slowest speed possible. The Steering wheel was rotated to the maximum that it can rotate during turning to the left and right directions. The Radii of Turning Circle and Turning Area with and without brakes were measured. The Radius of Turning Circle is the Radius of the smallest circle tangentially described by the median plane of the outermost wheel, while the Radius of Turning Area is the radius of the smallest circle described by the outermost point of the tractor. Results of the test are shown in **Tables 3** and **4**.

TURNING CIRCLE AND TURNING AREA

Tractor on Test : Mc CORMICK

Model : C100max

Date of Test : January 3, 2013

Table 3. The Radius of Turning Circle of the tractor

	RIGHT TURN	LEFT TURN
WITHOUT BRAKES	5863	5663
WITH BRAKES	4188	4200

Table 4. The Radius of Turning Area of the tractor

	RIGHT TURN	LEFT TURN
WITHOUT BRAKES	5958	5758
WITH BRAKES	4283	4295

B. DETERMINATION OF TRAVELLING SPEEDS

The Travelling Speeds of the tractor were measured in actual condition. These were measured on a 50-meter concrete road. The tractor was operated at full throttle speed setting and made to run at different range and speed settings. The time elapsed for the tractor to cross the 50-meter distance was obtained. The fourth gear setting at range "HIGH", yielded the fastest Forward Travelling speed of 31.4 kph, while the first gear setting at range "LOW" was slowest at 1.89 kph. For the Reverse Travelling Speed the fastest speed was 32.8 kph, while the slowest speed was 1.95 kph. In general, the tractor speeds increased with increased range and gear settings. **Table 5** shows the results of the test.

TRAVELLING SPEEDS

Tractor on Test	:	Mc CORMICK
Model	:	C100max
Date of Test	:	January 3, 2013

Table 5. The Travelling Speeds of the Tractor in kph at different settings

Main Gear Setting	SHUTTLE and RANGE SETTINGS SPEED (kph)					
	FORWARD			REVERSE		
	Low	Medium	High	Low	Medium	High
1	1.89	4.51	11.1	1.95	4.58	11.2
2	2.97	7.07	17.3	3.02	6.63	17.9
3	3.74	8.93	21.8	3.80	8.89	22.1
4	5.55	13.0	31.4	5.59	13.2	32.8

C. PTO PERFORMANCE LOAD TEST

The power at the Power-Take-Off shaft of the tractor was obtained using a Portable Waterbrake Dynamometer. It was tested at full throttle speed setting of the tractor. Data such as dynamometer load, PTO rpm, Engine rpm, and fuel consumption were obtained at 20 rpm speed increment. The Maximum No-Load speed of the engine was 2351 rpm. The Maximum PTO Output Power obtained was 61.5 kW at 601 rpm PTO shaft speed. At this point, the PTO torque was 99.6 kg-m with tractor's Fuel Consumption Rate of 17.9 L/h. No breakdown and other malfunctioning occurred during the test. The test set-up is shown in **Figure 10** and the results of test are shown in **Table 6** and plotted in **Figure 11**.

Figure 10. The Tractor on the PTO test set-up

PTO PERFORMANCE TEST

Tractor on Test : Mc CORMICK

Model : C100max

Date of Test : February 5, 2013

Table 6. Results of PTO load performance test

SHAFT SPEED (rpm)		PTO		FUEL CONSUMPTION (L/h)	SPECIFIC FUEL CONSUMPTION (g/kW-h)
ENGINE	PTO	TORQUE (kg-m)	POWER (kW)		
2351	654	17.4	11.7	8.70	627.0
2282	640	78.7	51.7	17.0	275.4
2224	622	95.4	60.7	17.8	246.9
2176	601	99.6	61.5	17.9	244.9
2079	581	101.8	60.7	17.8	246.7
2009	557	104.8	60.0	17.7	248.2
1949	539	107.7	59.5	17.5	246.9
1876	520	111.2	59.3	17.2	242.7
1751	500	115.4	59.2	16.7	236.9
1640	456	119.7	56.0	16.7	250.0
1467	411	125.3	52.8	16.0	253.8

REMARKS / OBSERVATIONS: No breakdown and any malfunctioning occurred.

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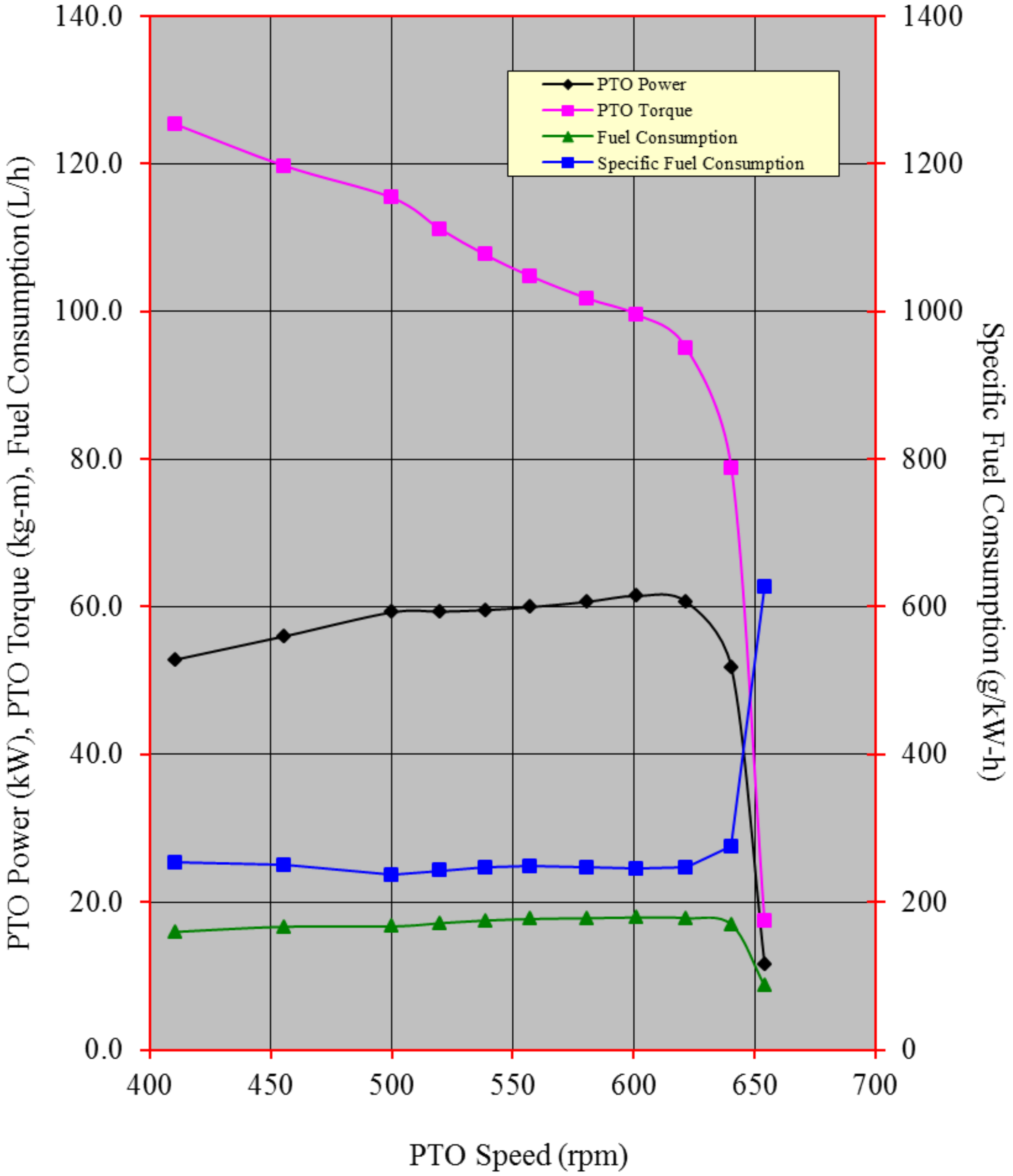

Figure 11. Performance Characteristic Curves of the Mc CORMICK C100max Four-Wheel Tractor

Test Engineer:

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ROMULO E. EUSEBIO

Checked by:

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DARWIN C. ARANGUREN

Approved for release:

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DELFIN C. SUMINISTRADO
Director

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Date

N.B.

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